Telecom Customer Churn Analysis and Prediction

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Abstract

Customer churn poses a significant challenge in the telecommunications industry, where acquiring a new customer is substantially more expensive than retaining an existing one. This project presents an end-to-end machine learning pipeline for churn prediction using the IBM Telco Customer Churn dataset (7,032 records). We perform comprehensive data preprocessing, exploratory data analysis, and feature engineering, then train and evaluate three classification models: Logistic Regression, Decision Tree, and Random Forest. Class imbalance is addressed using SMOTE. All models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, with recall emphasized as the primary metric. Results show that Logistic Regression and Random Forest achieve AUC ≈ 0.817, with churn recall of 64% and 65% respectively. Feature importance analysis consistently identifies payment method, tenure, contract type, and monthly charges as the strongest predictors of churn.

1 Introduction

Customer churn—the rate at which customers stop doing business with a company—is a critical metric in subscription-based industries. In telecommunications, switching costs are low and competition is high, making retention a constant challenge. Retaining existing customers is generally more cost-effective than acquiring new customers.

This project investigates the following research question:

> *Which customer features are most associated with telecom customer churn, and how effectively can machine learning models predict which customers are likely to leave?*

We address this through a full data science pipeline: preprocessing, exploratory data analysis (EDA), feature engineering, model training, and evaluation. Beyond predictive accuracy, we emphasize interpretability: understanding *why* customers churn is as important as predicting *whether* they will.

1.1 Prior Work

Churn prediction has been studied extensively across telecom, e-commerce, and banking. Standard approaches compare Logistic Regression, Decision Trees, and Random Forest on the same Telco dataset used here. A consistent finding is that contract type, monthly charges, and tenure are the strongest predictors, which our results confirm. A central challenge is class imbalance: churners typically represent 20–30% of customers. SMOTE [2] is widely used to address this, and we adopt it here.

2 Data and Preprocessing

2.1 Dataset

We use the IBM Telco Customer Churn dataset [1], sourced from Kaggle. The dataset contains 7,043 customer records across 21 features, including demographic information (gender, age, partner, dependents), service subscriptions (phone, internet, streaming, security), billing details (contract type, payment method, monthly and total charges), and a binary churn label. After removing 11 rows with missing `TotalCharges` values, the working dataset contains **7,032 records**.

2.2 Preprocessing Pipeline

Feature removal. `customerID` is a unique identifier with no predictive value and was dropped.

Type conversion. `TotalCharges` was stored as a string in the raw CSV; it was coerced to numeric and the 11 resulting `NaN` rows were dropped.

Binary encoding. Columns with two categories (gender, Partner, Dependents, PhoneService, PaperlessBilling, Churn) were mapped to 0/1.

Internet-service encoding. Internet-related service columns (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, MultipleLines, InternetService) were mapped such that any active service becomes 1, while `No internet service` and `No phone service` become 0.

One-hot encoding. Contract (three values) and PaymentMethod (four values) were one-hot encoded, producing six binary dummy features.

Feature engineering. Two derived features were created:

- **totalServices**: count of active service subscriptions per customer (range 0–9).
- **riskFlag**: binary indicator set to 1 when `tenure` $<$ 37 months *and* `MonthlyCharges` $>$ \$61, identifying customers who are both new and high-paying.

Train/test split. Data was split 80/20 (5,625 training, 1,407 test) using stratified sampling. SMOTE was applied *only* to the training set, balancing churn and non-churn to 50/50 before model fitting.

3 Exploratory Data Analysis

3.1 Churn Distribution

The dataset is imbalanced: 26.6% of customers churned (1,869 records) and 73.4% stayed (5,163 records), as shown in Figure 1. This motivates the use of SMOTE and the prioritization of recall over accuracy as the primary evaluation metric.

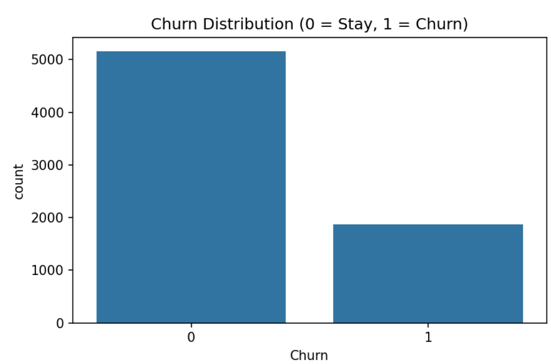

Figure 1: Churn distribution. Approximately 26.6% of customers churned, creating a class imbalance that must be addressed during model training.

3.2 Feature Correlations

Figure 2 shows the Pearson correlation of each feature with the churn label. **Tenure** has the strongest negative correlation ($r = -0.35$), meaning customers with longer history are substantially less likely to leave. Internet-related services and **MonthlyCharges** show positive correlation with churn, while **OnlineSecurity** ($r = -0.17$) and **TechSupport** ($r = -0.16$) show negative correlation, suggesting that support-related services improve retention.

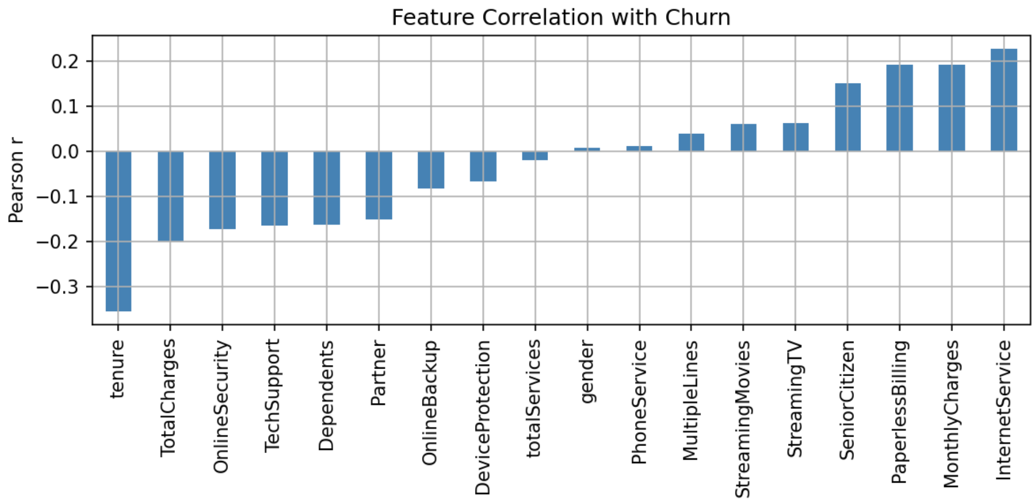

Figure 2: Pearson correlation of each feature with churn. Tenure is the strongest negative predictor; InternetService and MonthlyCharges are the strongest positive predictors.

3.3 Contract Type

Contract type is the most visually striking predictor (Figure 3). Month-to-month customers churn at **42.7%**, one-year contract customers at 11.3%, and two-year contract customers at just **2.8%**. The absence of a long-term commitment creates a low barrier to switching.

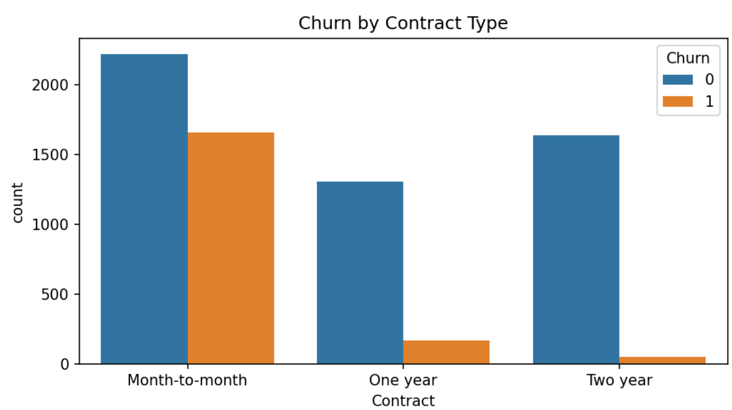

Figure 3: Churn counts by contract type. Month-to-month customers account for the vast majority of churners; two-year customers almost never leave.

3.4 Tenure and Monthly Charges

Churned customers average **18.0 months** of tenure, compared to **37.7 months** for customers who stayed (Figure 4). The scatter plot in Figure 5 shows that churners concentrate in the upper-left quadrant—short tenure combined with high monthly charges—which motivates the **riskFlag** feature. Among high-risk customers (riskFlag = 1), the churn rate is **51.3%**, versus only 16.2% for the rest.

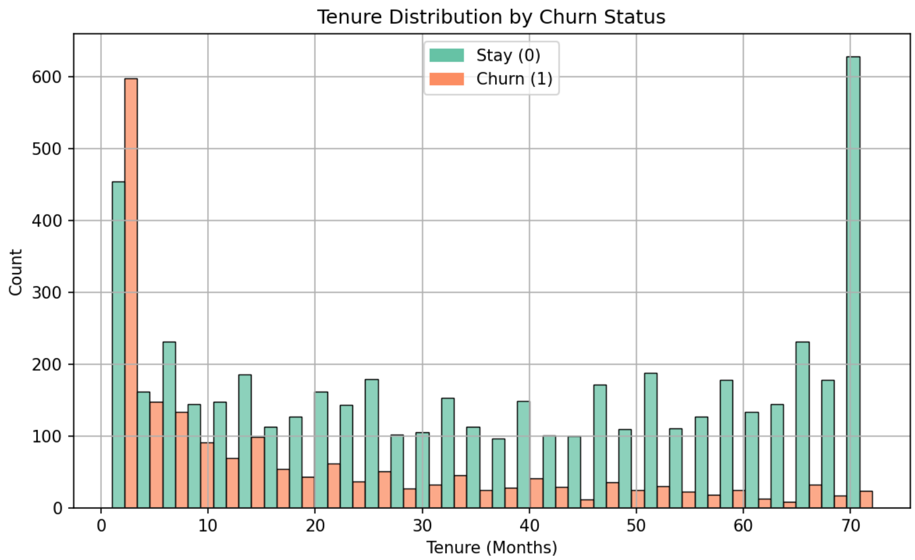

Figure 4: Tenure distribution by churn status. Churners spike in the first 1–5 months; loyal customers accumulate at 60–72 months.

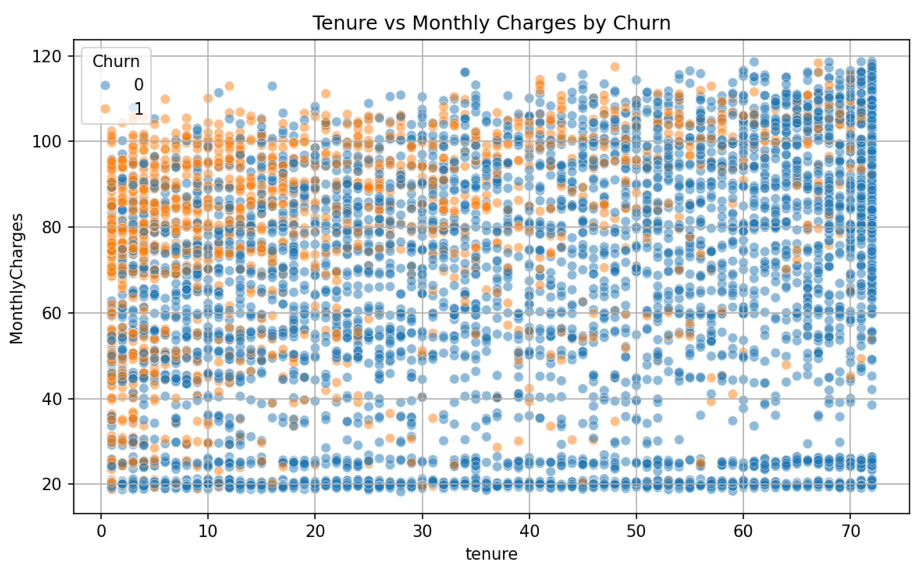

Figure 5: Tenure vs. Monthly Charges colored by churn. Churners cluster in the high-charge, short-tenure region, validating the riskFlag feature.

3.5 Correlation Heatmap

Figure 6 confirms expected multicollinearity: TotalCharges is highly correlated with both tenure (0.83) and MonthlyCharges (0.65). The engineered riskFlag correlates with Churn at $r = 0.36$, making it the second-strongest numerical predictor after tenure.

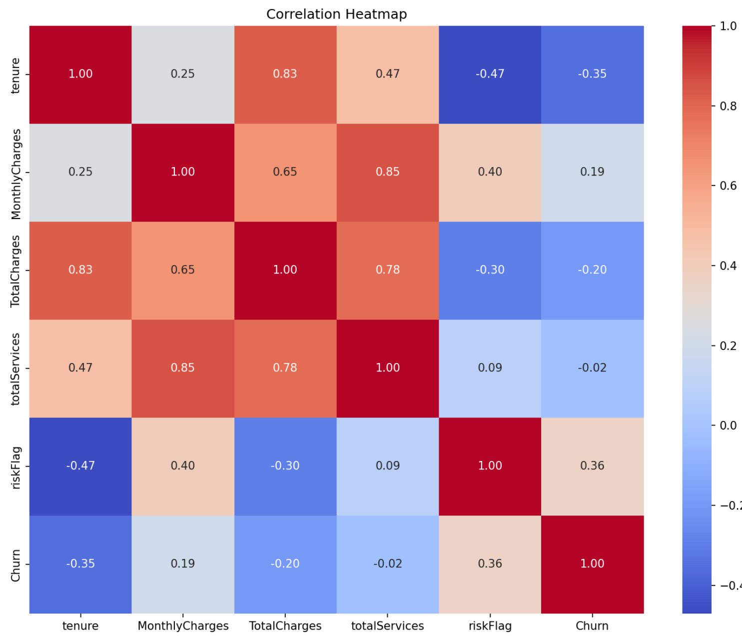

Figure 6: Correlation heatmap of numerical features. The strong TotalCharges–tenure correlation (0.83) reflects its composite nature. riskFlag achieves $r = 0.36$ with Churn.

4 Methodology

4.1 Models

Three classifiers were selected to span the complexity–interpretability tradeoff:

Logistic Regression. Used as the interpretable baseline. Churn probability is estimated via:

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\alpha + \beta^\top \mathbf{x})}}.$$

Parameters were estimated by maximum likelihood with `max_iter=3000`. Coefficients provide direct insight into each feature's directional effect on churn probability.

Decision Tree. Trained with `max_depth=6` to prevent overfitting. Splits maximize information gain. The resulting tree is interpretable as a set of human-readable rules; feature importances reflect aggregate impurity reduction.

Random Forest. An ensemble of 100 decision trees (`max_depth=10`) [3], each trained on a random feature and sample subset. Averaging predictions across trees reduces variance and improves generalization. Feature importances are averaged across all trees.

4.2 Evaluation Metrics

Accuracy alone is insufficient under class imbalance. We report:

- **Recall** (primary): fraction of actual churners correctly identified. Missing a churner is costly—the customer is lost entirely.
- **Precision**: fraction of predicted churners who actually churned. Low precision wastes retention resources.
- **F1-score**: harmonic mean of precision and recall.
- **ROC-AUC**: discriminative ability across all classification thresholds.
- **Confusion matrix**: breakdown of true/false positives and negatives.

5 Results

5.1 Model Performance

Table 1 summarizes performance on the held-out test set (1,407 samples). All three models achieve similar accuracy (74–76%); differences emerge most clearly in recall and ROC-AUC.

Table 1: Model performance on the test set (1,407 samples). Recall is the primary metric. Logistic Regression achieves the best balance of recall and ROC-AUC.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.760	0.54	0.64	0.59	**0.817**
Decision Tree	0.740	0.52	**0.66**	0.58	0.785
Random Forest	0.750	0.53	0.65	0.58	0.816

5.2 ROC-AUC Curves

Figure 7 shows ROC curves for all three models. Logistic Regression (AUC = 0.817) and Random Forest (AUC = 0.816) are nearly indistinguishable, both well above the diagonal. The Decision Tree (AUC = 0.785) trails slightly—a typical result, since individual trees are more sensitive to training-data variance than ensembles.

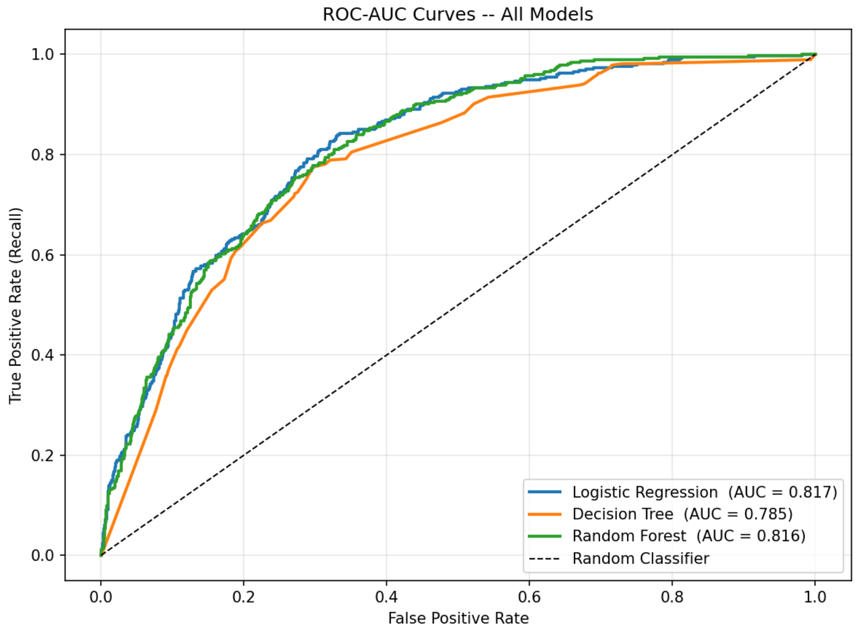

Figure 7: ROC-AUC curves for all three models. Logistic Regression and Random Forest achieve near-identical AUC of 0.817 and 0.816, both substantially above the random baseline.

5.3 Confusion Matrices

Figure 8 shows confusion matrices on the test set. Of 374 true churners: Logistic Regression correctly identifies 239, Decision Tree 248, and Random Forest 243. However, the Decision Tree also produces the most false positives (233), while Logistic Regression produces the fewest (203). This trade-off has a direct business interpretation: if the goal is to catch as many churners as possible, the Decision Tree wins on recall; if the goal is efficient targeting of retention resources, Logistic Regression's lower false-positive rate is preferable.

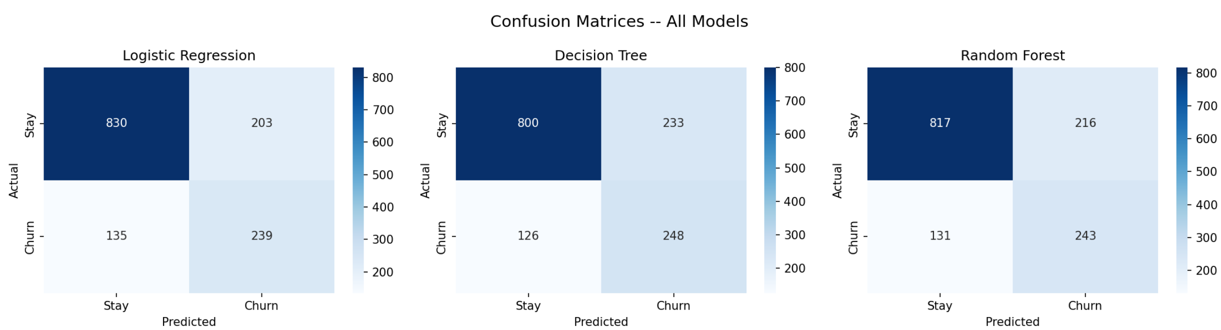

Figure 8: Confusion matrices for all three models. Decision Tree catches the most churners (248) at the cost of the most false positives (233). Logistic Regression offers the best precision–recall balance.

5.4 Feature Importance

Figure 9 shows feature importances across all three models. Key findings:

- **PaymentMethod_Electronic check** consistently appears among the strongest predictors across all models, with importance 0.44 in Decision Tree and 0.15 in Random Forest, and shows a large positive coefficient in Logistic Regression. Customers paying by electronic check churn at a higher rate, possibly reflecting lower engagement or fewer automatic payment commitments.

- **Tenure** is the dominant predictor by Logistic Regression coefficient magnitude (most negative: longer tenure strongly predicts staying) and ranks second in Random Forest.
- **MonthlyCharges** ranks third in Random Forest, consistent with its positive EDA correlation with churn.
- **Contract type** (two-year, one-year) ranks in the top features of all three models, confirming that contractual commitment is a primary retention mechanism.
- **OnlineSecurity** and **TechSupport** rank highly in Logistic Regression and Random Forest, suggesting that value-added services reduce churn risk.

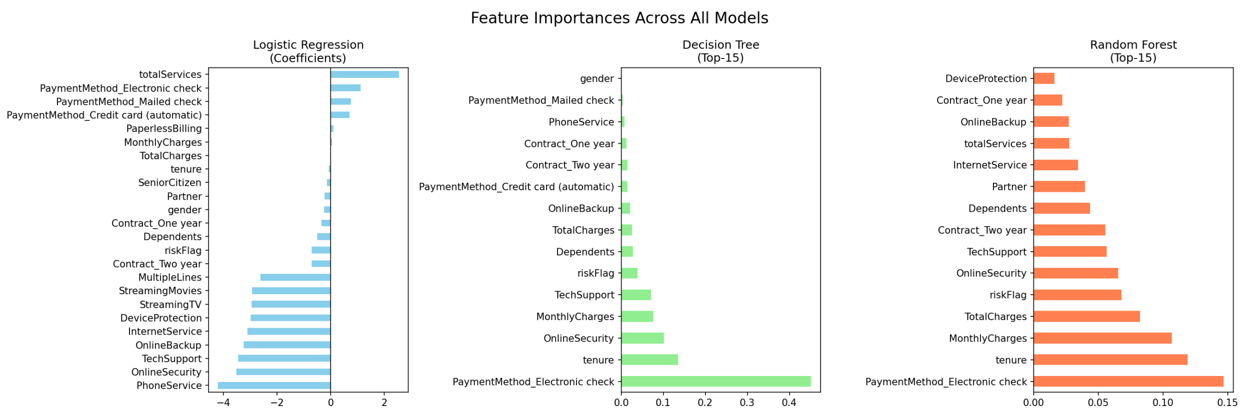

Figure 9: Feature importances across all three models. Despite different algorithms, all three consistently identify payment method, tenure, and contract type as dominant predictors.

6 Discussion

The results provide insight into the research question. The features most associated with churn are: (1) electronic check payment, (2) short tenure, (3) month-to-month contract, (4) high monthly charges, and (5) absence of security or support add-ons. This profile describes a customer who is new, pays a premium, has no long-term commitment, and lacks additional support or security-related services—more likely to churn based on observed patterns.

All three models achieve AUC ≈ 0.82, representing strong lift over random guessing. The engineered **riskFlag** feature demonstrates that domain-driven feature engineering can surface high-risk segments without complex modeling. Its 51.3% churn rate among flagged customers (vs. 16.2% for others) makes it useful for identifying high-risk customer groups.

Limitations. The dataset covers a single provider, so generalizability to other markets is not guaranteed. SMOTE was applied before feature importance computation, meaning importance scores partly reflect the augmented distribution. No hyperparameter tuning was performed; grid or random search could improve recall further, particularly for Random Forest.

7 Conclusion

This project demonstrated that telecom customer churn can be predicted with meaningful accuracy using demographic and billing features. Key takeaways:

- Tenure, contract type, and payment method are the strongest predictors of churn.
- Month-to-month customers churn at 42.7%—fifteen times the rate of two-year contract customers (2.8%).
- Logistic Regression achieves AUC = 0.817 and recall = 64%, matching the more complex Random Forest (AUC = 0.816, recall = 65%).
- A simple riskFlag feature identifies a segment churning at 51.3%.

The most actionable recommendation is to prioritize retention outreach toward month-to-month customers paying by electronic check with tenure under 37 months, particularly those without online security or tech support. Contract upgrade incentives and bundled service discounts targeted at this segment could meaningfully reduce churn.

References

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[2] Chawla, N.V., Bowyer, K.W., Hall, L.O., & Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

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[4] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.